Mathematics - Graduate Level Assessment 3 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Miranda enlarged a picture proportionally. Her original picture is 4 cm wide and 6 cm long. If the new, larger picture is 10 cm wide, what is its length?
   (a) 22 cm
   (b) 15 cm
   (c) 16 cm
   (d) 10 cm
   (e) 12 cm

2. For how many positive integers k does the ordinary decimal representation of the integer k! end in exactly 99 zeros?
   (a) Twenty
   (b) Three
   (c) None
   (d) Four
   (e) Five

3. Compute covariance of x=(1,2,3,4), y=(2,3,4,5)
   (a) 0.50
   (b) 2.00
   (c) 3.00
   (d) 0.75
   (e) 1.25

4. Find the sum of $\$\sum_{stv} b\infty ntwwmbmtufuhcvgefbqgaxsfevcrxxj$

    5. -2.459
    6. -1.459
    7. -0.159
    8. 2.459
    9. 0.959

Each time Rami turned the dial on a machine, the dial moved 1 degree. Rami turned the dial 10 times. What is the total number of degrees the dial moved?

1. 270
2. 100
3. 90
4. 20
5. 110

True / False (10 marks)

Answer True or False

6. True or False: The number of ways to arrange 6 pairs of parentheses such that they are balanced
7. True or False: The value of the series _{n=0}^{}(-1)^{n} rac{1}{3n+1} is approximately 1.1356
8. True or False: If each puzzle cost $6, the total cost of 934 puzzles sold would be $5,764.
9. True or False: The difference in elevation between Talon Bluff and Salt Flats is 695 feet.
10. True or False: (1+i)^{10} = 32i, demonstrating the application of De Moivre's theorem in complex

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Compute
$$\sum_{pe} cnn \log_2\left(1+\frac{s}{e}\right)\log_k 2\log_{am} eg$$
12. Discuss the various arrangements possible for a panel consisting of three students and six faculty members seated at a table. Provide a detailed analysis of the permutations involved.

End of Paper